

PLAYSPHERE ULTRA-COMPREHENSIVE MASTER ARCHITECTURE SPECIFICATION

Platform: Grassroots Sports OS for India (Telangana Focus)

Repository: PlaySphere (Flutter Mobile/Web + Firebase + Node.js Cloud Functions)

1. EXECUTIVE ARCHITECTURE PILLARS

1. OFFLINE-FIRST EVENT SOURCING (MUST):

Scoring happens on grounds with zero cellular signal. All actions append to an immutable MatchEvent log. Local device is the source of truth; sync is background.

2. DYNAMIC RULECONFIG ABSTRACTION (MUST):

No sport rules (overs, set caps, point values, raid time) are hardcoded. Every scoring engine reads dynamic RuleConfig overrides per competition.

3. SINGLE AUTHORIZED SCORER LOCK INVARIANT:

To prevent write conflicts, each match has exactly one active scorer lock.

4. ZERO-TRUST SECURITY BOUNDARY (firestore.rules):

Client handles UI; server firestore.rules enforces append-only logs, role-based authorization, and minor safety under the Indian DPDP Act.

5. SERVER-SIDE FINANCIAL & RATING LEDGERS (Cloud Functions asia-south1):

Glicko-2 ratings and official tournament ranking points are computed strictly in Cloud Functions to eliminate client-side rank tampering.

2. SYSTEM ARCHITECTURE & LAYER BREAKDOWN

[PRESENTATION LAYER] Flutter 3.x / Material 3 / GoRouter v14.2.0

- Declarative URLs: /home, /org/:id, /event/:id/score/:id, /watch/:id
- Responsive Layouts: Touch targets >= 60pt on mobile, keyboard shortcuts on laptop
- Localization: AppLocalizations (Telugu, Hindi, English resource files)

[STATE MANAGEMENT LAYER] Riverpod v2.5.1

- Providers: currentUserProvider, authStateProvider, fixtureProvider

[DOMAIN & SPORT ENGINE LAYER] Pure Functional Dart Modules

- Sport Engines: Cricket, Badminton, Football, Volleyball, Kabaddi, Kho-Kho
- AI Team Balancer: Snake draft + rating differential minimizer
- UNDO Engine: Reconstructs state from append-only MatchEvent log reversals

3. COMPLETE SCREEN-BY-SCREEN COMPONENT INVENTORY

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1. HomeScreen (/home) Member Launchpad
 - _HomeHeader: Welcome banner, profile avatar, search bar
 - _MyClubsSection: Carousel of joined clubs with role badges (Owner, Admin, Scorer)
 - _LiveMatchesBar: Active live matches across user's clubs with real-time score tickers
 - _QuickActionGrid: 1-tap shortcuts to Quick Match, Looking For Board, Officials
2. OrgHomeScreen (/org/:orgId) Club Dashboard
 - _ClubHeader: Logo, banner, location tag (Mandal/District), total members count
 - _ClubFeed: Announcements list, poll widgets, recent memory media grid
 - _UpcomingEventsSection: Event cards with 1-tap RSVP buttons (Going, Maybe, No)
 - _InterClubChallengesCard: Incoming challenges with Accept/Decline actions
3. QuickMatchScreen (/org/:orgId/quick-match) Casual Match Setup
 - DropdownButtonFormField: Sport selector (Cricket, Badminton, Kabaddi, etc.)
 - ChoiceChip Row: Side format selector (Singles, Doubles, 6-a-side, 11-a-side)
 - _RulesetCard: Expandable preset selector (T20, 8-over Gully) with editable rules
 - _SideCard: Side A & Side B roster list with 1-tap member toggles
 - _AutoBalanceButton: 1-tap AI Team Auto-Balance trigger using Glicko-2 ratings
 - OutlinedButton ('Add by code'): Add players from other clubs via PSOS code
 - FilledButton ('Start match & score'): Creates fixture and opens ScoringScreen
4. ScoringScreen (/org/:orgId/event/:compId/score/:fixtureId) Scoring Pad
 - _Scoreboard: High-contrast scoreboard card (teams, headline score, summary)
 - _PendingBanner: Offline queue badge showing pending actions waiting to upload
 - FilledButton.tonalIcon ('Undo last'): Always-visible 1-tap UNDO button
 - _ControlRow: Dynamic action buttons ([0],[1],[2],[4],[6],[Wd],[Nb],[Wicket])
 - MatchScorecard: Full box score tables (BattingLine, BowlingLine, FallOfWicket)
 - MatchMemoriesSection: In-match photo/video clip uploader and thumbnail grid
 - _AdminActions: Outcome overrides for walkover, retirement, disqualification

4. PUBLIC VIEWS, PROFILES & GOVERNMENT DASHBOARD

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- 5. SpectatorScreen (/org/:orgId/event/:compId/watch/:fixtureId) Public Live View
 - _LiveHeader: Real-time score ticker backed by Firestore snapshot listener
 - _WagonWheelWidget: Interactive graphic showing shot angles and distances
 - _BallByBallFeed: Chronological commentary timeline with event icons
 - 6. CareerProfileScreen (/player/:uid) Portable Lifelong Player Identity
 - _ProfileHeader: Photo, verification tier (Self, Scorer Verified, Association)
 - _CrossSportIndexCard: Overall 'Sports OS Index' score (0 100)
 - _RatingHistoryChart: Line chart tracking Glicko-2 rating progression
 - 7. AnalyticsScreen (/org/:orgId/analytics) B2G Government Dashboard
 - _FilterHeader: Location picker (State -> District -> Mandal -> Village)
 - _KpiMetricGrid: Active participants, match count, venue utilization %, para ratio

5. DATABASE SCHEMAS & ENTITY RELATIONSHIP MODEL

ENTITIES TABLE:

- users: uid (PK), phone, name, dateOfBirth, profileVisibility, verificationTier, district
- orgs: id (PK), name, type (community|school|college|village), location, inviteCode
- memberships: uid (PK), orgId (FK), role (owner|admin|scorer|member), status
- competitions: id (PK), orgId (FK), sportId, format, status, ruleConfig
- fixtures: id (PK), competitionId (FK), entrantAId, entrantBId, scorerUid, scoreState
- match_events: docId (000001 PK), fixtureId (FK), seq, type, payload, clientEventId
- tournaments: id (PK), orgId (FK), grade (club|district|state|national), status
- ranking_entries: id (PK), uid (FK), tournamentId (FK), points, round, awardedAt
- ratings: sportId (PK), uid (FK), rating (r), deviation (RD), volatility, settledFixtures

6. END-TO-END EVENT FLOWS & 5G RECONCILIATION

FLOW A: OFFLINE MATCH SCORING LIFECYCLE

1. Scorer Action: Tap action button ([4], [Wicket]) on ScoringScreen.
2. Pure Validation: SportEngine.validate() checks rules against RuleConfig.
3. Local Render (<100ms): State updates locally; event enqueued to SharedPreferences.
4. UI Badge: Shows 'X actions waiting to sync' banner.
5. 5G Reconnect: ScoringService.reconcileQueue() flushes batch to Cloud Firestore.
6. Triggers Fire: Glicko-2 ratings update + Multicast FCM push sent to club members.

FLOW B: 5G RECONCILIATION & QUEUE SAFEGUARDS

[Offline Ground Scoring] > [Write UUIDv7 to Queue] > [Instant Local Render]

(5G Connection Hits)

[Firestore SDK Flush] > [SyncBatchPlanner.plan()] > [Cap Batch at 500 Events]

[Monotonic Seq & Lock Check] > [Atomic Server Commit] > [Clear Pending Badge]

[Cloud Functions Trigger: Glicko-2 Ratings + Multicast FCM Push + Spectator Push]

7. CRITICAL PROTECTIONS ON RECONNECT:

- Idempotency: Every event docId is sequence padded string (events/0000042). No duplicate balls.
- Causal Order: Events replayed strictly in seq order (1 -> 2 -> 3).
- 500-Event Cap: Multi-day offline tournaments flush in 500-event steady passes.